

Economic Analysis Portfolio

Economic Analysis of Medication Adherence Improvement in Diabetes Care

Implementation Planning, Financial Analysis, and Organizational Impact

Maria Ruth Su, M.S., PMP

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Research Operations Portfolio

github.com/mariasu-pmp/research-operations-portfolio

Executive Summary

Medication nonadherence remains one of the most significant barriers to improving outcomes for patients with Type 2 diabetes mellitus (T2DM), contributing to preventable complications, increased healthcare utilization, and rising healthcare expenditures. This analysis evaluates the economic and operational feasibility of improving medication adherence through structured intervention strategies within an integrated healthcare system.

The analysis compares implementation considerations, financial impact, and organizational value before identifying a pharmacist-led medication adherence intervention as the preferred strategy. The proposed implementation plan incorporates standardized workflows, multidisciplinary coordination, patient education, and ongoing monitoring to improve adherence while supporting operational scalability.

Financial modeling estimates first-year program costs of approximately \$298,008, with projected break-even occurring within 12–18 months. Long-term projections suggest cumulative savings exceeding \$1.8 million over five years through reductions in hospitalizations, diabetes-related complications, and avoidable healthcare utilization. The analysis illustrates how structured implementation planning and continuous monitoring can support sustainable quality improvement under the modeled implementation scenario.

Beyond its financial evaluation, this project presents a structured approach to implementation planning, evidence synthesis, workflow design, and organizational decision-making. The methodology illustrates how economic analysis can inform operational strategy while balancing clinical outcomes, implementation feasibility, and long-term organizational value.

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Background

Medication adherence is one of the most significant determinants of health outcomes for patients with Type 2 diabetes mellitus (T2DM). Despite the availability of effective pharmacologic therapies, many patients do not take medications as prescribed, resulting in poorer glycemic control, increased rates of diabetes-related complications, and substantially higher healthcare expenditures. Improving medication adherence therefore represents both a clinical priority and an operational opportunity for healthcare organizations seeking to improve patient outcomes while reducing avoidable costs.

Integrated healthcare systems increasingly recognize medication adherence as a strategic quality improvement objective because preventable hospitalizations, emergency department utilization, and long-term complications contribute significantly to the overall cost of diabetes care. Evidence from the healthcare literature indicates that structured interventions to improve medication adherence can reduce healthcare utilization while improving clinical outcomes, making medication adherence an important area for operational and economic evaluation.

This analysis examines the economic and organizational implications of improving medication adherence in patients with Type 2 diabetes. Rather than evaluating clinical effectiveness alone, the project considers implementation feasibility, financial sustainability, workflow integration, and long-term organizational impact. Multiple intervention approaches were reviewed before identifying a pharmacist-led strategy as the preferred implementation model based on its combination of clinical effectiveness, operational feasibility, and projected economic value.

Problem Statement

Medication nonadherence remains a persistent challenge in diabetes management despite the availability of effective therapies. Approximately half of patients with chronic conditions do not consistently follow prescribed medication regimens, contributing to preventable disease progression, increased hospitalization rates, and higher healthcare expenditures. In diabetes care, poor adherence is associated with complications including cardiovascular disease, nephropathy, retinopathy, and reduced quality of life. At the same time, avoidable healthcare costs attributable to medication nonadherence continue to place significant financial pressure on healthcare systems.

For integrated healthcare organizations, improving medication adherence is more than a clinical objective—it is an operational and economic challenge that requires coordinated implementation, multidisciplinary collaboration, and sustainable workflow design. The central question addressed in this analysis is:

How can healthcare organizations improve medication adherence in patients with Type 2 diabetes while achieving financially sustainable and operationally scalable implementation?

To answer this question, the project evaluates implementation strategies, economic outcomes, and organizational considerations before recommending an evidence-based intervention designed to improve both patient outcomes and long-term organizational performance.

Methodology and Assumptions

This analysis evaluates the economic feasibility and organizational implications of improving medication adherence among patients with Type 2 diabetes mellitus through structured intervention strategies. Rather than conducting a primary economic study, the analysis integrates published clinical evidence, healthcare economics literature, and implementation planning to develop a financial model to support organizational decision-making.

The financial model is based on three categories of assumptions:

Evidence Base

- Published studies evaluating medication adherence interventions.
- Healthcare economics literature examining reductions in hospitalizations and cost savings associated with improved medication adherence.
- Evidence supporting pharmacist-led interventions in chronic disease management.

Financial Assumptions

- Personnel costs were estimated using the proposed staffing model.
- Fringe benefits were calculated at 30 percent.
- Direct implementation costs included technology, educational materials, and administrative resources.
- Projected savings were estimated using conservative assumptions derived from published healthcare literature.

Modeling Approach

Projected savings were estimated by applying published evidence describing the relationship between improved medication adherence and reductions in healthcare utilization to the proposed implementation model, using conservative assumptions to estimate long-term organizational impact. The analysis is intended to support organizational planning by illustrating the potential economic implications of implementing a pharmacist-led medication adherence program under the stated assumptions.

Intervention Analysis

Evaluating Strategies to Improve Medication Adherence

Multiple intervention approaches have been proposed to improve medication adherence among patients with Type 2 diabetes, including digital reminder systems, patient education programs, and pharmacist-led interventions. Each strategy offers distinct advantages but also differs in implementation complexity, operational requirements, scalability, and long-term sustainability.

This analysis evaluated several intervention approaches using three primary criteria:

- Clinical effectiveness
- Operational feasibility
- Economic value

Digital reminder systems offer relatively low implementation costs and can improve adherence among selected patient populations. However, their effectiveness depends heavily on sustained patient engagement and technology adoption. Educational interventions improve patient knowledge but often require ongoing reinforcement to produce lasting behavioral change.

Following a comparative evaluation, a pharmacist-led intervention was selected as the preferred implementation strategy. Published evidence suggests that pharmacists are well positioned to improve medication adherence through medication counseling, identification of adherence barriers, individualized patient education, and ongoing follow-up. Because pharmacists are already integrated within medication management workflows, the intervention can be incorporated into existing healthcare operations while minimizing additional infrastructure requirements.

From an operational perspective, the pharmacist-led model also provides greater opportunities for continuous quality improvement through standardized workflows, multidisciplinary collaboration, and ongoing performance monitoring. These characteristics support both long-term sustainability and organizational scalability.

Implementation Strategy

Operational Implementation Framework

The recommended intervention was designed as a structured implementation program that integrates pharmacist expertise with existing pharmacy workflows. The implementation strategy emphasizes standardized processes, multidisciplinary coordination, and continuous performance monitoring to improve medication adherence while maintaining operational efficiency.

Implementation consists of six coordinated phases:

1. Develop Standardized Intervention Workflow

Establish a pharmacist-led workflow incorporating medication adherence assessments, documentation standards, follow-up protocols, and quality monitoring procedures.

2. Identify Target Patients

Generate patient outreach lists using prescription refill data and clinical records to identify individuals with demonstrated adherence gaps.

3. Coordinate Patient Consultations

Schedule pharmacist consultations promptly after patient identification, coordinating appointment logistics with pharmacy staff.

4. Deliver Individualized Medication Counseling

Provide patient-centered education addressing medication adherence barriers, diabetes self-management, and individualized intervention strategies.

5. Maintain Accessible Service Delivery

Support both in-person and virtual consultation models to improve patient participation while reducing barriers to engagement.

6. Monitor Performance and Continuous Improvement

Track adherence outcomes through ongoing performance measurement, allowing intervention strategies to be refined over time using continuous quality improvement principles.

The structured implementation approach was selected because it balances clinical effectiveness with operational feasibility while supporting long-term organizational scalability.

Economic Analysis

Financial Evaluation

The proposed medication adherence program was evaluated from an organizational perspective to assess whether the anticipated improvements in patient outcomes justify the required implementation investment. This analysis considers implementation costs, projected healthcare savings, return on investment (ROI), and long-term financial sustainability using a financial model informed by published literature and the assumptions described in the Methodology and Assumptions section.

Implementation Investment

The proposed implementation requires both personnel and direct operational investments. Personnel costs are driven primarily by pharmacist-led patient education and follow-up activities and include two pharmacists (25% FTE each), four pharmacy technicians (25% FTE each), and one pharmacy manager (10% FTE). Including fringe benefits, total projected personnel costs are estimated at **\$215,508**. Direct implementation costs, including computers, checkout devices, educational materials, and administrative resources, are estimated at **\$82,500**, resulting in an estimated first-year implementation cost of approximately \$298,008.

Projected Financial Benefits

Under the assumptions of the financial model, improvements in medication adherence are projected to reduce hospitalizations, emergency department utilization, and diabetes-related complications. Based on published evidence incorporated into the model, projected first-year healthcare savings are estimated at approximately **\$250,000–\$300,000**, with financial projections indicating break-even within **12–18 months**. Long-term modeling suggests cumulative savings exceeding **\$1.8 million** over five years through reductions in avoidable healthcare utilization and chronic disease complications.

Return on Investment

Under the model assumptions, first-year ROI is projected at approximately **–8%**, reflecting initial implementation costs during program startup. As adherence improvements become sustained and implementation costs are offset by reductions in healthcare utilization, **the financial model projects ROI increasing to approximately 450% over five years**. These estimates are intended to illustrate the potential long-term economic value of the proposed intervention under the stated assumptions rather than predict guaranteed implementation outcomes.

Organizational Impact

Operational Value Beyond Financial Performance

While financial outcomes are central to implementation decisions, the organizational value of medication adherence programs extends beyond measurable cost savings.

Improved medication adherence supports organizational objectives by reducing preventable hospitalizations, improving chronic disease management, strengthening patient engagement, and promoting more effective use of healthcare resources. Structured implementation also enhances multidisciplinary collaboration by integrating pharmacists, pharmacy technicians, and clinical teams into coordinated adherence workflows.

From an operational perspective, the proposed intervention establishes standardized documentation practices, defined implementation workflows, and measurable performance indicators that support

continuous organizational learning. These characteristics improve implementation consistency while enabling future quality improvement initiatives to build upon established operational processes.

The intervention was also evaluated for scalability. Because the implementation relies primarily on existing pharmacy infrastructure and multidisciplinary collaboration rather than specialized technology, the model can be adapted across integrated healthcare systems and community health settings. Similar implementation strategies may also be extended to other chronic disease populations, including hypertension and hyperlipidemia, where medication adherence remains a persistent challenge.

Limitations

This analysis is intended to support organizational decision-making through structured economic evaluation rather than predict precise financial outcomes. The projections presented are based on published literature, conservative financial assumptions, and a modeled implementation scenario. Actual implementation outcomes may vary depending on patient population characteristics, organizational workflows, intervention fidelity, and local healthcare costs.

In addition, the analysis evaluates a pharmacist-led intervention within the context of an integrated healthcare system. Organizations operating under different reimbursement models, staffing structures, or patient populations may experience different implementation costs and economic outcomes. Consequently, the financial estimates presented should be interpreted as scenario-based projections intended to inform planning rather than guaranteed operational results.

Key Takeaways

1. Medication adherence is both a clinical priority and an organizational investment opportunity.
2. Successful implementation depends on integrating clinical effectiveness with operational feasibility, financial sustainability, and standardized workflows.
3. Pharmacist-led interventions provide a practical model for improving medication adherence while leveraging existing healthcare infrastructure.
4. Economic evaluation supports evidence-based organizational decision-making by helping healthcare leaders assess the feasibility of implementation before investing in quality improvement initiatives.
5. The analytical framework presented here can be adapted to evaluate implementation strategies for other chronic disease management programs.

Portfolio Reflection

This analysis originated as graduate coursework in healthcare economics and has been substantially revised for publication as part of my professional Economic Analysis Portfolio. The portfolio version emphasizes implementation planning, economic evaluation, organizational decision-making, and evidence-based analysis for a professional audience while preserving the analytical framework and conclusions of the original work.

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